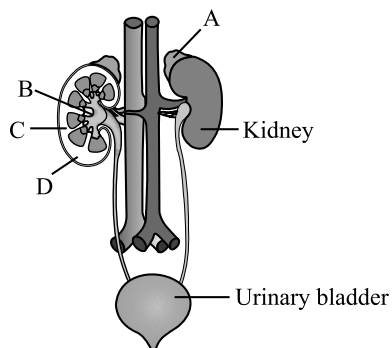


CHAPTER 16

Excretory Products and their Elimination

Excretory Organs, Human Excretory System

- Choose the correct statement given below regarding juxta medullary nephron. (2024)
 - Loop of Henle of juxta medullary nephron runs deep into medulla.
 - Juxta medullary nephrons outnumber the cortical nephrons.
 - Juxta medullary nephrons are located in the columns of Bertini.
 - Renal corpuscle of juxta medullary nephron lies in the outer portion of the renal medulla.
- Given below are statements: one is labelled as Assertion A and the other is labelled as Reason R. (2023)
Assertion (A): Nephrons are of two types: Cortical & Juxta medullary, based on their relative position in cortex and medulla.
Reason (R): Juxta medullary nephrons have short loop of Henle whereas, cortical nephrons have longer loop of Henle.
 In the light of the above statements, choose the correct answer from the options given below:
 - A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is NOT the correct explanation of A.
 - A is true but R is false.
- Nitrogenous waste is excreted in the form of pellet or paste by (2022)
 - Pavo*
 - Ornithorhynchus*
 - Salamandra*
 - Hippocampus*
- In mammals, which blood vessel would normally carry largest amount of urea? (2016 - I)
 - Renal Vein
 - Dorsal Aorta
 - Hepatic Vein
 - Hepatic Portal Vein
- Figure shows human urinary system with structures labeled A to D. Select option which correctly identifies them and gives their characteristics and/ or functions: (2013)



- D-Cortex - Outer part of kidney and do not contain any part of nephrons.
- A-Adrenal gland- Located at the anterior part of kidney. Secrete catecholamines which stimulate glycogen breakdown.
- B-Pelvis-Broad funnel shaped space inner to hilum, directly connected to loops of Henle's.
- C-Medulla - Inner zone of kidney and contains complete nephrons.

Urine Formation, Function of Tubules

- Given below are two statements: (2024)
Statement-I: In the nephron, the descending limb of loop of Henle impermeable to water and permeable to electrolytes.
Statement-II: The proximal convoluted tubule is lined by simple columnar brush border epithelium and increases the surface area for reabsorption.
 In the light of the above statements, choose the correct answer from the options given below :
 - Statement-I is true but Statement-II is false
 - Statement-I is false but Statement-II is true
 - Both Statement-I and Statement-II are true
 - Both Statement-I and Statement-II are false
- Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**
Assertion (A): Ascending limb of loop of Henle is impermeable to water and allows transport of electrolytes actively or passively.
Reason (R): Dilution of filtrate takes place due to efflux of electrolytes in the medullary fluid.
 In the light of the above statements, choose the **correct** answer from the options given below: (2023-Manipur)
 - (A) is true but (R) is false
 - (A) is false but (R) is true
 - Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Both (A) and (R) are true and (R) is not the correct explanation of (A)

8. Which of the following statements is correct?

(2017-Delhi)

- The ascending limb of loop of Henle is impermeable to water
- The descending limb of loop of Henle is impermeable to water
- The ascending limb of loop of Henle is permeable to water
- The descending limb of loop of Henle is permeable to electrolytes

9. The part of nephron involved in active reabsorption of sodium is:

(2016 - II)

- Bowman's capsule
- Descending limb of Henle's loop
- Distal convoluted tubule
- Proximal convoluted tubule

10. Removal of proximal convoluted tubule from the nephron will result in:

(2015)

- No change in quality and quantity of urine
- No urine formation
- More diluted urine
- More concentrated urine

11. Which of the following does not favour the formation of large quantities of dilute urine?

(2015)

- Renin
- Atrial-natriuretic factor
- Alcohol
- Caffeine

Mechanism of Concentration of Filtrate, Regulation of Kidney Functions, Micturition

12. Diuresis is prevented by:

(2024 Re)

- Renin from JG cells via switching off the osmoreceptors
- ANF from atria of the heart
- Aldosterone from adrenal medulla
- Vasopressin from Neurohypophysis

13. Given below are two statements:

(2024 Re)

Statement-I: Concentrated urine is formed due to counter current mechanism in nephron.

Statement-II: Counter current mechanism helps to maintain osmotic gradient in the medullary interstitium.

In the light of the above statements, choose the most appropriate answer from the options given below.

- Statement-I is correct but Statement-II is incorrect.
- Statement-I is incorrect but Statement-II is correct.
- Both Statement-I and Statement-II are correct.
- Both Statement-I and Statement-II are incorrect.

14. Which of the following statements are correct?

(2023)

- An excessive loss of body fluid from the body switches off osmoreceptors.
- ADH facilitates water reabsorption to prevent diuresis.
- ANF causes vasodilation.

D. ADH causes increase in blood pressure.

E. ADH is responsible for decrease in GFR.

Choose the correct answer from the options given below:

- C, D and E only
- A and B only
- B, C and D only
- A, B and E only

15. Arrange the events of Renin – Angiotensin mechanism in correct sequence.

(2023-Manipur)

- Activation of JG cells and release of renin.
- Angiotensin II activates release of aldosterone.
- Fall in glomerular blood pressure.
- Reabsorption of Na^+ and water from distal convoluted tubule.
- Angiotensinogen is converted to Angiotensin I and then to Angiotensin II.

Choose the correct answer from the options given below:

- (C), (A), (E), (B), (D)
- (A), (D), (E), (C), (B)
- (A), (D), (C), (B), (E)
- (B), (A), (E), (D), (C)

16. Select the correct statements.

(2022 Re)

- Angiotensin II activates the cortex of adrenal gland to release aldosterone.
- Aldosterone leads to increase in blood pressure.
- ANF acts as a check on renin-angiotensin mechanism.
- ADH causes vasodilation.
- Vasopressin is released from adenohypophysis.

Choose the most appropriate answer from the options given below:

- (A), (B) and (C) only
- (A), (B) and (E) only
- (C), (D) and (E) only
- (B), (C) and (D) only

17. Select the correct statement:

(2020-Covid)

- Angiotensin II is a powerful vasodilator.
- Counter current pattern of blood flow is not observed in vasa recta.
- Reduction in Glomerular Filtration Rate activates JG cells to release renin.
- Atrial Natriuretic Factor increases the blood pressure.

18. The increase in osmolarity from outer to inner medullary interstitium is maintained due to:

(2020-Covid)

- Close proximity between Henle's loop and vasa recta
 - Counter current mechanism
 - Selective secretion of HCO_3^- and hydrogen ions in PCT
 - Higher blood pressure in glomerular capillaries
- (iii) and (iv)
 - (i), (ii) and (iii)
 - (i) and (ii)
 - Only (ii)

19. Which of the following factors is responsible for the formation of concentrated urine?

(2019)

- Low levels of antidiuretic hormone
- Maintaining hyperosmolarity towards inner medullary interstitium in the kidneys.
- Secretion of erythropoietin by Juxtaglomerular complex
- Hydrostatic pressure during glomerular filtration

20. Match the items given in Column-I with those Column-II and select the correct option given below (2018)

	Column-I (Function)		Column-II (Part of Excretory System)
A.	Ultrafiltration	i.	Henle's loop
B.	Concentration of urine	ii.	Ureter
C.	Transport of urine	iii.	Urinary bladder
D.	Storage of urine	iv.	Malpighian corpuscle
		v.	Proximal convoluted tubule

- a. A-iv B-v C-ii D-iii
 b. A-iv B-i C-ii D-iii
 c. A-v B-iv C-i D-ii
 d. A-v B-iv C-i D-iii

21. A decrease in blood pressure/volume will **not** cause the release of (2017-Delhi)
 a. Renin
 b. Atrial Natriuretic Factor
 c. Aldosterone
 d. ADH
22. Which of the following does not favour the formation of large quantities of dilute urine? (2015)
 a. Renin
 b. Atrial-natriuretic factor
 c. Alcohol
 d. Caffeine
23. Human urine is usually acidic because: (2015 Re)
 a. Excreted plasma proteins are acidic
 b. Potassium and sodium exchange generates acidity
 c. Hydrogen ions are actively secreted into the filtrate
 d. The sodium transporter exchanges one hydrogen ion for each sodium ion, in peritubular capillaries

24. Which of the following causes an increase in sodium reabsorption in the distal convoluted tubule? (2014)

- a. Decrease in antidiuretic hormone levels
 b. Increase in aldosterone levels
 c. Increase in antidiuretic hormone levels
 d. Decrease in aldosterone levels

Role of Other Organs and Excretory Disorders

25. Use of an artificial kidney during hemodialysis may result in: (2019)

- A. Nitrogenous waste build-up in the body
 B. Non-elimination of excess potassium ions
 C. Reduced absorption of calcium ions from gastro-intestinal tract
 D. Reduced RBC production

Which of the following option is the most appropriate?

- a. (A) and (B) are correct b. (B) and (C) are correct
 c. (C) and (D) are correct d. (A) and (D) are correct

26. Match the items given in Column-I with those in Column-II and select the correct option given below (2018)

	Column-I		Column-II
A.	Glycosuria	i.	Accumulation of uric acid in joints
B.	Gout	ii.	Mass of crystallised salts within the kidney
C.	Renal calculi	iii.	Inflammation in glomeruli
D.	Glomerular nephritis	iv.	Presence of glucose in urine

- a. A-iii B-ii C-iv D-i
 b. A-i B-ii C-iii D-iv
 c. A-ii B-iii C-i D-iv
 d. A-iv B-i C-ii D-iii

Answer Key

1. (a) 2. (d) 3. (a) 4. (c) 5. (b) 6. (d) 7. (c) 8. (a) 9. (d) 10. (c)
 11. (a) 12. (d) 13. (c) 14. (c) 15. (a) 16. (a) 17. (c) 18. (c) 19. (b) 20. (b)
 21. (b) 22. (a) 23. (c) 24. (b) 25. (c) 26. (d)